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Hormonal changes, semen quality and variance in reproductive activity outcomes of post pubertal rabbits fed *Moringa oleifera* Lam. leaf powder

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ABSTRACT

Ethnopharmacological relevance: Moringa oleifera Lam. (Moringaceae) is an important plant based staple food, known for its nutritional and medicinal value and is usually prescribed by herbal practitioners in Nigeria and in other tropical countries for the treatment of male infertility problems and reproductive diseases in females. Although the aphrodisiac properties and fertility enhancement potential in males have been reported, the underlying mechanisms for the activity remain unclear. In this study, we investigated the influence of supplementing the diet with *M. oleifera* Lam. leaf powder on reproductive hormones and semen quality of New Zealand White (NZW) rabbits.

Materials and methods: Thirty-two (32) NZW rabbits of 50:50 ratio bucks to does, were randomly distributed to four treatment groups (n = 4 bucks, n = 4 does per group). Graded levels (0, 5, 10 and 15 g/kg) of *M. oleifera* Lam. leaf powder was incorporated into rabbit growers pellet. The does and bucks were housed separately in hutches and sheltered under the same environmental conditions with free access to their respective treatment diets for a period of 12 weeks.

Results: In female rabbits, treatment revealed significant ($P < 0.05$) dose-dependent reduction in the concentration of serum FSH, LH and oestrogen. While in contrast the highest dose of leaf powder significantly ($P < 0.05$) increased progesterone and prolactin concentrations remained unaffected. On the other hand, the concentration of FSH and LH in bucks was significantly ($P < 0.05$) increased in treatment groups compared to the control group. Serum testosterone concentrations were significantly lower in the 5 and 10 g/kg treatment groups. Semen volume, sperm count and motility were significantly improved in a dose dependent manner with increasing amounts of *M. oleifera* Lam. leaf powder in the diet.

Conclusions: We conclude that *M. oleifera* Lam. leaf powder supplementation to the diet was more beneficial to male rabbit fertility than the female, where it tended to have a negative impact through the hypothalamic-pituitary-gonadal axis. However, with the varying impact of *M. oleifera* Lam. leaf powder on the hypothalamic-pituitary-gonadal axis of male and female animals, further investigation is necessary to determine the mechanism through which it operates.

1. Introduction

Mammalian reproductive physiology is primarily regulated by the gonadotrophins luteinizing hormone (LH) and follicle stimulating hormone (FSH) secreted from the anterior pituitary which act on the gonads to produce sex steroids. The gonadotrophins are controlled by gonadotrophin releasing hormone (GnRH) secreted in a pulsatile manner from the hypothalamus. Thus, reproduction is tightly controlled by this hypothalamic-pituitary-gonadal axis. Interruption of these processes, in any of the functional events in either sex, leads to fertility impairment (Tsutsumi and Webster, 2009). Infertility and low fertility is a serious problem in both animals (domestic, companion or wild) and man. In farm animals, infertility is detrimental to production efficiency and the economics of food production (Donald, 1973) and in humans, assisted reproductive technologies have a limited success rate and are expensive. Herbal therapy that has actions on the hypothalamic-pituitary-gonadal axis may influence reproductive physiology and ameliorate some infertility problems (Wang et al., 2014; Ried,

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2015). Thus; identification of efficacious herbal plants that influence the hypothalamic-pituitary-gonadal axis may provide alternative treatments for infertility or as a contraceptive and perhaps ultimately lead to more effective medicines. A number of recent studies have identified plants that may have fertility enhancement (Lans et al., 2018) and psychopharmacological properties improving sexual behaviour (Kotta et al., 2013; Singh et al., 2013). For example, *Eurycoma longifolia* has been reported to boost serum testosterone concentration, have pro-fertility effects in animals (Chan et al., 2009) and man (Tambi et al., 2012) and has aphrodisiac properties (Asiah et al., 2007).

Moringa oleifera Lam. is one of the plants that is traditionally used by many communities in Nigeria and Sub Sahara African used both as food and medicine (Thurber and Fahey, 2009). It is used by the traditional medicine practitioners in Nigeria to enhance male fertility and in the treatment of reproductive diseases in women. It is a deciduous tree with sparse foliage and an average height of 5–10 m (Morton, 1991) and an aborigine of sub-Himalayan regions of Northwest India and widely distributed throughout rain forest/humid tropical and subtropical areas of Africa, Saudi Arabia, Southeast Asia, the Caribbean Islands, and South America (Afolabi et al., 2013; Ramachandran et al., 1980; Shukla et al., 1988).

M. oleifera Lam. belongs to the *Moringaceae* family, (synonym: M. ptreygosperma Gaertn) (Nadkarni, 1976), popularly called drumstick tree, horse radish tree, or kelor tree (Nadkarni, 1976; Anwar et al., 2003). It is a small deciduous plant, often planted in compounds or used as live fences in Nigeria (Aminu et al., 2011) and in other tropical countries where they thrive. *M. oleifera* leaves, fruit, flowers and immature pods are used as highly nutritive vegetables which can be eaten fresh, cooked, or stored as dried powder for many months without refrigeration, and without loss of nutritional value (Fahey, 2005). The leaves have been reported as an excellent source of protein, calcium, potassium, iron, vitamin, carotenoids and natural antioxidants and have been recommended as suitable for utilization in many developing countries where inadequate nourishment is of major concern (Fahey et al., 2001).

A recent review (Falowo et al., 2018) of phytochemical investigations of the roots pods and seeds, leaves and flowers of the plant has revealed the presence of a large range of compounds including antioxidants and nutraceuticals. *Moringa* leaves are also reported to be potent source of polyphenols, including quercetin-3-glycoside, rutin, kaempferol glycosides, and other polyphenols (Ndong et al., 2007) and antioxidant compounds such as ascorbic acid, flavonoids, phenolics and carotenoids (Dillard and German, 2000; Siddhuraju and Becker, 2003).

Every part of the tree (root, bark, gum, leaf, fruit pods, flowers, seed and seed oil) has been reported to have wide ranging medicinal properties (Gopalakrishnan et al., 2016; Falowo et al., 2018), such as antioxidant, antimicrobial, antitumor, anti-inflammatory cardioprotective activities as well as treatments for metabolic disorders such as type 2 diabetes. It is also used as an abortifacient, treatment for menstrual pain, aphrodisiac and as a fertility enhancement (Padmarao et al., 1996; Anwar et al., 2005; Nwamarah et al., 2015).

Aqueous extracts of *Moringa oliefera* seeds increased sexual behaviour (mounting frequency, ejaculation latency and intromission frequency) and significantly increased libido and sperm count in male mice (Zade et al., 2013). Leaf extracts added to the diet have been reported to increase epididymis and seminal vesicle weights in mice, but with no observable effect on LH and FSH (Cajuday and Pocsidio, 2010). More recently, Amera (2016) observed an improvement in fertility, hatchability percentage and egg production, and egg quality parameters in laying chickens. Similarly, *M. oleifera* have been reported to have fertility enhancement potentials such as aphrodisiac properties, increased fertilizing ability, enhanced mating behaviour, mounting behaviour, treatments of reproductive diseases in males (Ghosh et al., 2002; Kujo, 2004; Cajuday and Pocsidio, 2010; Akunna et al., 2012; Obembe et al., 2015; Dafaalla et al., 2017). Antifertility effect was reported by Prakash et al. (1987), Nath et al. (1992) and Farooq et al. (2012). However, it is not clear if the reported antifertility effect is associated with the reproductive endocrine system. Also, the link between *M. oleifera* and sex hormonal status and semen quality has not been substantiated.

Rabbits are an important livestock animal in Nigeria and other countries and maintaining and improving their fecundity is important to provide a sustainable food supply. Moringa *oleifera* leaf powder is a year-round renewable food source and potentially a useful additive to improve the nutritional quality of the rabbit feed and perhaps also the fertility of the animals. Therefore, this study was designed to assess to what extent *M. oleifera* Lam. leaf powder can influence basal concentrations of the reproductive hormones (LH, FSH, 17β-estradiol, progesterone testosterone, and prolactin) in males and female rabbits, as well as the semen quality in rabbit bucks.

2. Materials and method

2.1. Animals and housing

Male and female adult New Zealand White rabbits between 8 and 9 weeks of age were used in this study. The animals were bred and maintained in the rabbitry unit, Faculty of Agriculture Teaching and Demonstration farm, University of Port Harcourt, Nigeria and kept in the same litter mates with an average weight of 2.31 kg. They were maintained on commercial rabbit pellets offered ad libitum prior to the experiment. The experiment and all the procedures were approved by the research and ethics committee of the Faculty of Agriculture, University of Port Harcourt, Nigeria.

Ethical approval for this study was obtained from the ethics committee of the University of Port Harcourt, Choba, Nigeria and the study conformed to the NIH standard of care for experimental animals and good research practice.

2.2. Collection and preparation of plant material

Whole fresh leaves of *M. oleifera* Lam. were harvested at commercial farm (Alex Farm Ltd) located in Igwuruta, Rivers State, Nigeria in the South-South Region (Niger Delta) of Nigeria at longitude 4°57′15″N and latitude 7°0′45″E. The location is in the rain forest belt with warm and humid climate, and daily temperature ranging from 20 °C to 33 °C. The plants identity was confirmed by Dr. H.M. Ijeomah of the Department of Forestry and Wildlife Management, University of Port Harcourt, Nigeria. A voucher specimen (UPH-015) was deposited at the forestry herbarium in the Department of Forestry and Wildlife Management, University of Port Harcourt, Nigeria. The stalk was removed and the leaves sun dried and then pulverized to make a powder leaf meal.

2.3. Experimental design

To assess the effect of adding *M. oleifera* Lam. leaf powder to the diet on the reproductive hormone profile of male (n = 16) and female (n = 16) rabbits as well as semen characteristics of the bucks, New Zealand White rabbits of 50:50 ratio of bucks to does were randomly divided into four experimental groups viz: 0, 5, 10 and 15 g of *M. oleifera* Lam. leaf powder per kg of commercial rabbit growers pellet feed (Top Feeds Nigeria Limited, Sapele, Edo State, Nigeria). Each treatment group comprised of eight animals (n = 4 bucks, n = 4 does). There were no significant differences ($P < 0.05$) between ages, and weights of the animals in the treatment groups before the commencement of the experiment. The does had been monitored for regular oestrous cycles by vaginal cytology before being included in the study. The animals were housed individually in standard-type single tier hutches. The males were housed separately from the females in similar hutches and sheltered in the same environmental conditions and managed under the same conditions of husbandry as they had prior to the experiment which included regular washing and disinfection of

Table 1
The effects of increasing amounts of *M. oleifera* Lam. leaf powder in the diet on the reproductive hormone profile of rabbit does.

	Control	Treatment: g/kg *M. oleifera* Lam. leaf powder in diet		
		5	10	15
FSH (*IU/ml*)	2.700 ± 0.258[a]	1.075 ± 0.0957[ab]	0.855 ± 0.098[ab]	0.550 ± 0.100[b]
LH (*mIU/ml*)	3.225 ± 0.359[a]	2.425 ± 0.435[ab]	1.550 ± 0.208[ab]	0.925 ± 0.222[b]
PRO (*ng/ml*)	0.300 ± 0.0356[a]	0.575 ± 0.0957[ab]	0.500 ± 0.082[ab]	2.100 ± 0.560[b]
E2 (*pg/ml*)	88.000 ± 5.100[a]	25.75 ± 5.91[ab]	8.63 ± 2.360[b]	8.100 ± 1.431[b]
PRL (*ng/ml*)	1.350 ± 0.173	1.150 ± 0.1915	1.025 ± 0.163	1.000 ± 0.206

Data are expressed as mean ± std, n = 4. Within each hormone significantly different values are indicated with different superscripts.

feeding and drinking troughs. Water and the experimental diets were provided ad-libitum for the period of 12 weeks.

2.4. Blood collection and hormonal assays

After the expiration of the experimental period (12 weeks), the animals were sacrificed by cervical dislocation and blood samples collected from all the animals through cardiac puncture. The blood was allowed to clot at room temperature, and then centrifuged at 2500 rpm and the serum aspirated off and stored frozen until assayed.

The serum was assayed for testosterone, 17β-estradiol (E2), progesterone, FSH, LH and prolactin using commercial ELISA kits (AccuBind®, Monobind Inc. Lake Forest, CA 92630, USA).

2.5. Semen collection and analysis

The semen was collected using the method described by Herbert and Adejumo (1996) as modified by Ajuogu et al. (2015). Briefly, a reinforced polyvinyl chloride (PVC) tube with an inner diameter of 2.8 cm and an outer diameter of 3.8 cm was cut to a length of 3.5 cm (The larger tube - LT). Rigid plumbing hose with an inner diameter of 1.7 cm and an outer diameter of 2.7 cm cut to the same length representing the smaller tube (ST). The ST was inserted firmly into one end of the LT until about 2.5 cm of the former was allowed a total of 6.0 cm for the main part of the Artificial Vagina (AV). The junction between the two tubes ST and LT was sealed with super glue. Gold circle condom was used as a liner. The liner was inserted into the tube assembled from the top (large end) and overturned on the upper rim of the main AV unit and lightly held in place with a rubber band 5.0 cm in diameter and the other end subsequently pulled through the narrow end of the bottom tube and held. Glycerol (anti freezer and lubricator) 5–6 ml was poured into the space between the held liner and the tube till it was $^2/_3$ full. A 15-*ml* collecting tube (4.5 cm) inserted at the ST end from the bottom constituted the collecting vessels.

2.6. Ejaculation procedure/semen characteristics

Before use, the AV unit was warmed by placing it in a warm water bath of about 40–60 °C for 10–15 min according to Herbert and Adejumo (1996). A mature, non-gravid doe was used as a teaser. As the buck mounts and makes a thrust on the teaser doe before intromission, the penis was grabbed and the AV quickly applied from the side to the erect penis of the buck. The deep thrust of the penis into the AV elicits ejaculation within few seconds. Immediately after collection, the samples were analysed for semen quality characteristics. Microscope slides were prepared from the ejaculate samples. The slides were prepared and stained according to the method described by Kondracki et al. (2006). In each slide the morphological structure of 500 spermatozoa was evaluated, specifying the number of morphologically normal sperm and the number with morphological abnormalities, and distinguishing forms with major and minor defects according to the classification by Blom (1981). The pH value, volume, sperm motility, morphology, percentage of dead spermatozoa, percentage of sperm abnormalities, sperm-cell concentration, total sperms in the ejaculate, number of motile sperms per ejaculate were determined. Semen volume (*ml*) was measured as the volume of semen collected in one ejaculation; sperm count was measured as the number of sperm per millilitre (ml) of semen in one ejaculation; sperm morphology (%) was determined as the index of percentage sperm that had a normal shape; sperm motility (%) was determined as the percentage of sperm that have normal movement, and pH was measured as the degree of acidity (low pH) or alkalinity (high pH) of the semen.

2.7. Statistical analysis

The data was analysed using the SAS computer software package (SAS Institute Inc., Cary, NC, USA). The nonparametric Kruskal-Wallis test (Kruskal and Wallis, 1952) was used followed by a multiple comparison post hoc test using the SAS macro developed by Elliott and Hynan (2011). Data are presented and expressed as means ± standard deviation and unless otherwise stated $p < 0.05$ was considered significant.

3. Results

3.1. Reproductive hormones in rabbit does

Table 1 describes the serum concentration of FSH, LH, PRO, E2 and PRL of adult does fed *M. oleifera* leaf powder supplemented diet at graded levels of 0, 5, 10 and 15 g/kg. Prolactin was unaffected by the increasing amounts of leaf powder. The gonadotropins FSH and LH on the other hand significantly ($p < 0.05$) decreased in a dose dependent manner with those animals receiving the highest dose having a concentration only 20% FSH and 29% LH of the concentration of the control group. Estradiol was even more profoundly affected with the concentration declining to less than 10% of that of control levels at the 10 g/kg dose of *M. oleifera* leaf powder. In contrast to the other hormones serum progesterone concentrations although slightly elevated at the lower doses of leaf powder was significantly higher at the highest dose.

3.2. Reproductive hormones in rabbit bucks

The influence of *M. oleifera* leaf powder on the reproductive hormonal profile of rabbit bucks following 12 weeks of the experimental diet is shown in Table 2. Adding leaf powder to the diet of the rabbits significantly ($P < 0.02$) increased FSH concentrations over control. FSH was highest with the 0.5% leaf powder diet which then declined with the higher amounts of leaf powder diets such that they lower but not significantly from the 5 g/kg diet and higher than the control diet. Significant differences in LH concentrations between the diets was observed ($P < 0.001$), however the highest levels were found in the animals on the 10 g/kg leaf powder diet with lesser levels in both the 5 and 15 g/kg diets. Testosterone concentrations were significantly ($P < 0.02$) different across the excremental groups however was it was lowest in the 5 g/kg diet animals. Those animals on the 10 g/kg leaf

Table 2
Influence of *M. oleifera* Lam. leaf powder on the reproductive hormonal profile of rabbit bucks.

	Treatment: g/kg *M. oleifera* Lam. leaf powder in diet			
	Control	5	10	15
FSH (IU/ml)	0.825 ± 0.263[a]	2.425 ± 0.486[b]	1.575 ± 0.386[ab]	1.875 ± 0.628[ab]
LH (mIU/ml)	0.500 ± 0.163[a]	3.700 ± 1.055[ab]	5.300 ± 1.675[b]	2.575 ± 0.896[ab]
TST (nmol/L)	16.050 ± 1.482[a]	12.775 ± 1.028 [b]	13.975 ± 1.034 [ab]	16.075 ± 1.394 [a]

Data are expressed as mean ± std, n = 4. Within each hormone significantly different values are indicated with different superscripts.

Table 3
Effect of *M. oleifera* Lam. leaf powder on the semen quality characteristics of rabbit bucks.

	Treatment: g/kg *M. oleifera* Lam. leaf powder in diet			
Semen Parameter	**Control**	**5**	**10**	**15**
Semen volume (ml)	2.13 ± 0.30[a]	3.58 ± 0.60[ab]	3.75 ± 0.44[ab]	4.80 ± 0.36[b]
Normal morphology %	76.25 ± 4.80	87.50 ± 2.89	86.25 ± 4.79	83.75 ± 6.29
Sperm count (×10/ml)	587.5 ± 75.0[a]	762.5 ± 62.9[ab]	750.0 ± 70.7[ab]	825.0 ± 28.9[b]
Pus cells	1–2per/h	3–4per/h	1–2per/h	2–3per/h
Abnormal morphology (%)	23.75 ± 4.79	12.50 ± 2.89	13.75 ± 4.79	15.00 ± 7.07
Actively motile (%)	73.75 ± 4.79[a]	87.50 ± 6.45[ab]	80.00 ± 4.08[ab]	90.00 ± 4.08[b]
Sluggishly motile (%)	12.50 ± 2.89	7.50 ± 2.89	12.50 ± 5.00	6.25 ± 2.50
Dead cells	8.75 ± 4.79	5.00 ± 4.08	8.75 ± 2.50	3.75 ± 2.50

Data are expressed as mean ± std, n = 4. Within each hormone significantly different values are indicated with different superscripts.

powder diets were also low but not significantly ($P > 0.5$) different from either the control or other treatment groups. At the highest doses of leaf powder testosterone concentrations were not different from controls.

3.3. Semen quality characteristics

The effect of *M. oleifera* Lam. leaf powder on semen quality characteristics is presented in Table 3. Semen pH (8.0) and viscosity was normal in all groups (data not shown). No significant differences between experimental groups in morphology, the percentage of sluggishly motile sperm and the numbers of dead cells were observed. Semen volume and sperm count increased, while abnormal morphology decreased with increasing amounts of *M. oleifera* Lam. leaf powder in the diet. Increasing amounts of *M. oleifera* Lam. leaf powder significantly increased semen volume ($p < 0.01$) with the volume at the highest dose being 225% of the mean volume of the control rabbits. Sperm count also increased significantly ($p < 0.02$) with the sperm concentration being 40% higher at the highest dose of *M. oleifera* Lam. leaf powder, which increased the total sperm numbers in the ejaculate of the rabbits on the highest does of leaf powder to 317% of the sperm numbers in the ejaculate of the control rabbits. Interestingly the percentage of actively motile sperm increased significantly ($p < 0.02$) with increasing amounts of *M. oleifera* Lam. leaf powder in the diet.

4. Discussion

There have been a number of studies suggesting that various *M. oleifera* preparations have all manner of effects on reproductive physiology in a range of species including humans and rabbits. However, while these studies have reported outcomes few have attempted to investigate the mechanisms. Therefore, this study was designed using rabbits as animal model to assess impact of adding to the diet a graded dose of *M. oleifera* Lam. leaf powder on male and female fertility by determining the effects on the major reproductive hormones of hypothalamic pituitary gonadal axis. It has been suggested the hormones of the hypothalamic pituitary gonadal axis are essential reproductive biomarkers for the investigation of fertility and the functionality of reproduction in female animals (Harris and Naftolin, 1970). Additionally, the effects of the leaf powder on semen characteristics of the male rabbits was also investigated.

This study provides evidence that the anti-fertility effect of *M. oleifera* is a least in part due to its impact on the reproductive axis in rabbit does. The results of this study showed that FSH, LH and 17β-estradiol concentrations significantly decreased with increasing levels of *M. oleifera* Lam. leaf powder in a dose dependent pattern. Progesterone concentrations on the other hand rose in response to increasing amounts of leaf powder in the diet. The reduced concentrations of LH and FSH would lead to impairment of the growth and maturation of ovarian follicles (Martin, 1984; Popat et al., 2008). This conclusion is supported by the rapid decline of 17β estradiol concentrations with increasing doses of *M. oleifera* Lam. leaf powder. A reduction in LH and FSH with increasing doses of an ethanolic extract of *M. oleifera* leaves was similarly observed by Obediah and Paago (2018) in rats. Aqueous extracts of the seeds have also been shown to disrupt the estrous cycle in rats, which the authors attributed to the phytoestrogenic compounds (Zade et al., 2013) in the plant. Phytoestrogens are structurally similar to estradiol and can interact with oestrogen receptors to limit oestrous responses (Jefferson et al., 2009, 2005; Moutsatsou, 2007) and have been found in various extracts from *M. oleifera* including the leaves (Kasolo et al., 2011; Makita et al., 2016; Aminu et al., 2011; Ndong et al., 2007). These phytoestrogen compounds in other botanicals have been implicated in infertility problems in different animal models (Mazur, 1998; Dabhadkar and Zade, 2012; Jourdehi, 2014), and are very similar to other reports on the negative impact of medicinal plants on female reproductive physiology (Monsefi et al., 2006; Thakur et al., 2009; Asuquo et al., 2013).

Although, the exact mechanism behind the decrease in serum gonadotropins and oestrogen concentration seen in the female rabbits treated with the *M. oleifera* Lam. leaf powder is unclear, it is likely that the phytoestrogen compounds in the leaves acted as negative feedback at the hypothalamus and pituitary to attenuate GnRH and consequently LH and FSH concentrations. Joshi and Joshi (1992) reported that anti fertility herbs especially those used traditionally as contraceptives interferes with pituitary gonadotropin secretions which affects the functionality of ovary by directly inhibiting ovulation, implantation and gestation.

The compounds in the leaf powder that caused the observed dose-dependent reduction in the key reproductive hormones (FSH, LH and oestradiol) may also be due, at least in part to any of the other phytochemicals found in *M. oleifera* leaves including flavonoids, glucosinolates, anthocyanins, proanthocyanidins and cinnamates all of which have been reported as effective agents in plants used for abortion (Nath et al., 1992, 1997; Amaglo et al., 2010; Coppin et al., 2013; Torres-Castillo et al., 2013). These compounds may be acting alone or in synergism with each other to negatively influence reproductive function by significantly reducing FSH, LH and oestrogen concentrations observed in this study. Prakash et al. (1987) demonstrated that aqueous extracts of the root and bark of *M. oleifera* reduced pregnancy rates in rats and induced foetal resorption in late pregnancy. In another study that analysed anti-reproductive potential of folk medicinal plants, *M. oleifera* leaf extract was found to be 100% abortive with doses equivalent to 175 mg/kg of starting dry material (Nath et al., 1992). Similar findings have been reported for other plant extracts, Thakur et al. (2009), reported anti-fertility effect of aqueous and ethanolic extract of rhizome of *Curcuma longa* and seeds of *Carum carvi* and reduced LH, FSH and oestrogen concentrations were observed in female Wister rats treated with leaf extract of *Spondias mombin* (Asuquo et al., 2013). Therefore, the progressive reductions in oestrogen concentrations with increasing concentrations of leaf powder strongly suggest that ovarian antral follicle growth was reduced most likely due to the reduced concentrations of FSH induced by the rising concentrations of phytoestrogen compounds but we cannot rule out direct effects on the ovary.

The dose related increase in progesterone concentration observed in this study may suggest that *M. oleifera* leaf powder may be a useful tool in pregnancy support, improving mammary gland development and milk production, which are the cardinal physiological functions of progesterone in female animals (McNeilly et al., 1983; Mainka and Lthrop, 1990). Progesterone is synthesized from cholesterol via pregnenolone in the granulosa and theca cells of maturing follicles but primarily in the corpus luteum which is formed from the newly ovulated follicle under the influence of LH. It regulates several reproductive functions such as maintenance of pregnancy, development of mammary glands, suppression of uterine motility etc. (Nihnobu, 1985; Tuckey, 2005). The significant rise in progesterone concentrations in response to increasing doses of the leaf powder was surprising but suggests that the compounds in the *M. oleifera* may act directly on the uterus and ovary interfering in the release of prostaglandin-F2α resulting in persistent corpus lutea (Niswender et al., 2000).

The administration of graded levels of *M. oleifera* Lam. leaf powder showed and equally graded improvement in the concentration of LH and FSH concentrations and sperm numbers and motility in bucks. Our results may imply that *M. oleifera* Lam. leaf powder enhance sexual activity and capacity in rabbit bucks. Dafaalla et al. (2017) recently observed dose dependent increased serum concentration of FSH, LH and TST in male Albino rats administered with ethanolic extract of *M. oleifera* seed for 30 days. In contrast, Cajuday and Pocsidio (2010) reported no significant impact in serum LH and FSH concentration in male mice fed alcoholic extracts of *M. oleifera* leaves, however they did observe increased weights of the testis, epididymis and seminal vesicles. This finding would suggest that the increased sperm numbers and semen volume observed in this study was due to increases in size of the reproductive organs. This observed effect in this study and the increased weights of reproductive organs observed in mice (Cajuday and Pocsidio, 2010) may be related to the numerous antioxidant compounds found in *M. oleifera* Lam. leaf powder which were reported to heal oxidative stress-related testicular impairments in male animal (Ghosh et al., 2002; Kujo, 2004). Phyto-chemical investigations have shown that *M. oleifera* leaves are potent source of polyphenols, including quercetin-3-glycoside, rutin, kaempferol glycosides, and other polyphenols (Ndong et al., 2007) and antioxidant compounds such as ascorbic acid, flavonoids, phenolics and carotenoids (Dillard and German, 2000; Siddhuraju and Becker, 2003).

The bucks sex drive is controlled by the pulsatile release of hypothalamic gonadotropin-releasing hormone (GnRH) which regulates FSH and LH secretions from the anterior pituitary where it binds to its target cells in the testis, and increases the expression of steroidogenic acute regulatory protein (StAR) that stimulates the secretion of testosterone (McLachlan, 2000; Khisti et al., 2003; Lucki et al., 2012). It is likely that the components of the test plant maintain the pulse episodes of GnRH, leading to the observed elevated level of serum FSH and LH. However, the mechanism of action that differentiates the limiting action of the of *M. oleifera* Lam. leaf powder on the hypothalamic-pituitary-gonadal axis of the does with the enhancement of hypothalamic-pituitary-gonadal axis in the bucks remain unclear and require further investigation.

Administration of *M. oleifera* Lam. leaf powder which significantly increased the semen volume, sperm count, and motility clearly show that *M. oleifera* Lam. leaf powder enhances male fertility and that its phyto-compounds have the potential to improve semen quality characteristics. Our finding corroborates previous reports in rats (Akunna et al., 2012; Obembe et al., 2015) which demonstrated significant increases in sperm count supporting the increased weights of reproductive organs in mice (Cajuday and Pocsidio, 2010). This and previous studies suggest that the improved FSH and LH concentration may be a pointer to the fact that *M. oleifera* Lam. leaf powder phytocompounds may be exerting its effect through the hypothalamic pituitary gonadal (HPG) axis (Altınterim, 2014).

Apart from the reported activities of anti-oxidants which enhances reproductive function in males (Ghosh et al., 2002; Kujo, 2004), the observed positive influence of the leaf powder on bucks' reproductive function, may be related to the highly digestible nutritional profile of *M. oleifera* leaves with composition of high neutral detergent fibre; acid detergent fibre; crude protein; gross energy; and amino acids profile (Rubanza et al., 2005; Leone et al., 2015). *M. oleifera* leaves are also rich in carotenoids, potassium, calcium, vitamins (particularly C and E), and iron (Jongrungruangchok et al., 2010; Moyo et al., 2011; Yameogo et al., 2011). It is reported to contain essential and non-essential amino acids such as aspartic acid, alanine, threonine, glutamic acid, valine, glycine, isoleucine, leucine, lysine, histidine, tryptophan, phenylalanine, methionine and cysteine (Ganatra et al., 2012; Kesharwani et al., 2014; Leone et al., 2015). It is also an established rich source of omega-6, polyunsaturated fatty acids (PUFAs), such as linolenic acid, and linoleic acid, Palmitic acid, is recorded in the major saturated fatty acid, accounting for 16–18% of the total fatty acids in the *M. oleifera* leaves (Saini et al., 2014).

The fertility enhancing capacity of some selected herbal plant and its extracts has been documented in several studies. The popular Chinese and Indian perennial plant *Tribulus terrestris* Linn was observed to improve plasma testosterone, spermatogenesis and luteinizing hormone in male lamb and ram (Koumanov et al., 1982; Georgiev et al., 1988). Mullaicharam et al. (2004) reported aphrodisiac effect of male Wister rats fed of *Allium sativum* extracts at dosage rates of 0.57, 1.13 and 2.25 ml/kg, p.o. for 28 day and observed increased sexual behaviour in dose dependent manner. Ratnasooriya and Dharmasiri (2000) worked on the aphrodisiac potential of *Terminalia catappa* seeds at a dose of 1500 mg/kg or 3000 mg/kg, for 7 d in male rats and observed marked improvement of aphrodisiac action and sexual vigour. Suresh et al. (2010) assessed the ethanolic extracts of *Mucuna prurient's* seed at dose level of 200 mg/kg on sexual activity of male rats and observed significant and sustained increase in sexual activity, increased mounting frequency, intromission frequency, and ejaculation latency.

5. Conclusion

This study supports previous work in other species that the anti-fertility activities of *M. oleifera* Lam. leaf powder may be due to phytoestrogenic attenuation of the pituitary-gonadal axis that is evidenced by the reduced serum gonadotropin (FSH and LH) and oestrogen concentration in does. The progesterone concentration which progressively increased in a dose-dependent manner may suggest that *M. oleifera*

Lam. leaf powder could be useful to support pregnancy, improve mammary gland development and milk production. The results reported here provide evidence that *M. oleifera* Lam. leaf powder could be used to enhance bucks reproductive function clearly manifested through the improvement of the endocrine profile and semen quality characteristics. However, with the varied impact of *M. oleifera* Lam. leaf powder on the hypothalamic-pituitary-gonadal axis of male and female animals, further investigation is necessary to determine the mechanism through which it operates. Therefore, it is safe to conclude that the *M. oleifera* Lam. leaf powder is more beneficial to male rabbit reproductive function than the female where it may lead to infertility.

Authors' note

Peter K. Ajuogu (pajuogu@myune.edu.au) conceived and designed the study, participated in data analysis and results interpretation, prepared the initial draft of the manuscript, and participated in the critical review and revision of the article. *Osaro Mgbere* (omgbere@uh.edu) conducted data analysis, interpreted the results, participated in the preparation of the initial draft of the manuscript, and subsequent critical review and revision of the article. *Disere S. Bila* (diserebila@gmail.com) managed experimental animals, collected data and samples, participated in data analysis, results interpretation and critical review and revision of the article. *James McFarlane* (jmcfarla@une.edu.au) interpreted the study findings and participated in the critical review and revision of the article for important intellectual content in area of reproductive physiology and endocrinology. All authors read and approved the final version of the article for publication.

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